

Processing asymmetries and presupposition projection: the case of binary connectives

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Abstract

Presuppositions triggered in the left argument of binary connectives like *if* and *or* typically become presuppositions of the complex sentence. By contrast, presuppositions triggered in the right argument varyingly project conditionally or unconditionally depending on the context — a phenomenon often referred to as the ‘proviso problem’. For example, *If Phil travels to London, he will bring his guitar* presupposes unconditionally that Phil has a guitar. However, *If Phil is a musician, he will bring his guitar* presupposes this only if Phil is a musician. Theories diverge on how to account for this variation. Some propose that speakers accommodate stronger propositions than the conditional presuppositions directly derived from the semantics. Others argue that presuppositions are semantically unconditional but can be cancelled when incompatible with the context. Both views thus account for the data, but rely on fundamentally different mechanisms.

This paper investigates the underlying mechanism by leveraging the processing signatures of presupposition cancellation. When a sentence presupposes *p*, contexts incompatible with *p* should trigger cancellation of that presupposition, though sentences associated with a weaker presupposition ‘If *A*, then *p*’, need not. Across three experiments, we detected a ‘cancellation cost’ during processing when presuppositions were triggered in the left argument of conditionals and disjunctions, though not in the right argument. These results suggest that presuppositions in the right argument project in a weaker form compared to those in the left.

1 Introduction

A hallmark property of presuppositions that distinguishes them from asserted content that they *project*. Presuppositions initially introduced in an embedded clause often, though not invariably, get inherited by higher-level constructions in which these clauses are embedded. A major research agenda in linguistics and philosophy of language has been to identify the right “algorithm” for presupposition projection — one that can derive the presupposition of a complex sentence from its elementary parts in a compositionally transparent way.

A simple and intuitively appealing idea is that presuppositions project ‘cumulatively’. That is, presuppositions of constituents simply get inherited wholesale as presuppositions of the complex

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construction (Langendoen & Savin 1971). This works well for presuppositions embedded under negation. Both (1) and (2), for instance, have identical presuppositions: Phil has a guitar.

- (1) Phil's guitar is broken.
Presupposition: Phil has a guitar.
Asserted content: That guitar is broken.
- (2) It's not the case that Phil's guitar is broken.
Presupposition: Phil has a guitar.
Asserted content: That guitar is not broken.

But at least since Karttunen (1973), the cumulative hypothesis has been known to be descriptively inadequate. While some operators like negation pass up the embedded presupposition unchanged, others show more complex behavior. When a presupposition is triggered in the left argument of a binary connective like a conditional or a disjunction, it gets inherited wholesale, as in (3). But when a presupposition is triggered in the right argument, the projected presupposition is often of a weaker, conditionalized form.

- (3) a. If Phil's guitar is broken by now, he is a real musician.
Presupposes: Phil has a guitar.
 b. Either Phil's guitar is broken by now, or he isn't a real musician.
Presupposes: Phil has a guitar.
- (4) a. If Phil is a musician, he will bring his guitar to the party.
Presupposes: If Phil is a musician, he has a guitar.
 b. Either Phil is not a musician, or he will bring his guitar to the party.
Presupposes: If Phil is a musician, he has a guitar.

Karttunen (1974) introduced the notion of 'local' contexts to capture the differences in how presuppositions project from the left and right arguments of connectives. He suggested that utterances change the context in several consecutive steps, creating intermediate contexts that include information from earlier parts of the sentence. When evaluating a conditional such as (4), the antecedent would be evaluated relative to the global, utterance context c , whereas the consequent would be interpreted relative to a context c' that results from updating c with the antecedent. Presuppositions need to be satisfied in — i.e. entailed by — their *local* context. This means that presuppositions triggered in the antecedent must hold in all worlds of c , while those in the consequent need to hold only conditionally: they need to be true in c' , the subset of worlds in c in which the antecedent is also true.

Many theories of presuppositions since (Heim, 1983; Schlenker, 2008; Chemla and Schlenker, 2012; Fox, 2008; George, 2008b) incorporate Karttunen's core insight in one way or other and derive weaker, conditionalized presuppositions from the right argument of binary connectives. There are important conceptual and formal differences among these theories, but for expository convenience, we will refer to them as c -theories (' c ' for conditional)¹. What is most relevant for us is that on c -theories, presuppositions triggered in the left and right arguments of these binary connectives project differently. If A presupposes p , then "If A_p , B " also presupposes p ; but if B

¹We follow Mandelkern (2016) in this naming convention.

presupposes p , “If A , B_p ” does not automatically presuppose p , though it does presuppose *if* A , p .

But sometimes this prediction of conditional presuppositions is overly weak, as pointed out by van der Sandt (1992) and Geurts (1996). The presupposition instead seems to project unconditionally. To illustrate, consider (5). Both sentences in (5) have a presupposition triggered in the consequent of the conditional. Their presuppositions seem to differ, however: whereas the felt presupposition of (5-a) conforms to predictions of c -theories, we infer (5-b) to have a stronger presupposition than what is predicted — that Phil has a guitar *simpliciter*.

- (5) a. If Phil is a musician, he will bring his guitar to the party.
Presupposes: If Phil is a musician, he has a guitar.
b. If Phil is traveling to London, he will bring his guitar on the trip.
Presupposes: Phil has a guitar.

This mismatch between the predicted (weaker, conditional) presupposition and what is felt to be presupposed has come to be known as the “Proviso Problem”. From a single logical environment, namely the right argument of conditionals, we see two patterns of presupposition projection. Sometimes presuppositions triggered in the right argument become presuppositions of the conditional as is, but sometimes they project in a weaker conditionalized form. And this variability seems to be based on changes in surrounding lexical material, which typically are not thought to trigger shifts in logical behavior.

The response by c -theorists to the proviso problem is to maintain that the *semantic* presupposition is conditional, but supplement the semantics with a pragmatic strengthening mechanism that leads listeners to accommodate a stronger one (Stalnaker 1973, 1974; Karttunen 1974; Heim 1983, 1990, 1992; Beaver 2001; von Stechow 2008; van Rooij 2007; Singh 2007; Rothschild 2011; Kleene 1952; van Fraassen 1969; Peters 1979; Kracht 1994; Beaver & Krahmer 2001; George 2008b; Lassiter 2012, Lauer 2015, Sudo 2014; a.o.; Fox 2008, 2013). The proviso problem, on this view, reflects the fact that accommodation is not always minimal. Listeners may strengthen conditional presuppositions when the weaker presupposition is implausible in context. In the case of (5-b) above, the minimal conditional presupposition would suggest an implausible connection between traveling to London and owning a guitar. Instead, a more reasonable inference is that the speaker presupposes that Phil has a guitar *simpliciter*. Theories differ on what governs this strengthening, with proposals ranging from plausibility-based assessments of the conditional presupposition to the independence between the antecedent and the presupposition of the consequent. What connects these views is the idea that the felt presuppositions from the right arguments of binary connectives are not only judgments about their semantic presuppositions, but also reflect further inferences drawn when presented with presuppositional sentences out-of-the-blue.

An alternative response is to posit a semantic theory that generates basic unconditional presuppositions, supplemented by a mechanism that weakens these presuppositions to conditional ones in certain contexts (e.g., Gazdar 1979; van der Sandt 1992; Geurts 1996; Mandelkern 2016). We refer to proponents of this approach as u -theorists (u for “unconditional”). U -theories have been motivated by various challenges raised against strengthening-based accounts over the years (e.g., Geurts 1999, Mandelkern 2016), though attempts have been made to address these challenges (e.g., Beaver 2001; Heim 1983; Fox 2019).

A concrete implementation of this view can be found in Discourse Representation Theory

(DRT). In the variant developed by Geurts (1996, 1999), presuppositions, like pronouns, are either “bound” within a Discourse Representation Structure (DRS) or accommodated. A DRS is a formal structure containing discourse referents and conditions on them. When encountering a presupposition trigger, the hearer seeks an antecedent within the DRS, potentially resolving the presupposition locally and preventing it from projecting globally. Thus in a sentence like in (6), the presupposition that Phil has a guitar is bound in the antecedent, preventing it from projecting further and resulting in the sentence having no global presuppositions.

(6) If Phil has a guitar, his guitar is expensive.

When no antecedent is available, presuppositions must be accommodated. While DRT allows for multiple potential sites of accommodation, there are interpretive constraints that favor global accommodation whenever possible (see Geurts 1999, p. 59). Thus, the preferred reading for (7) is one where the sentence globally presupposes that Phil has a guitar.

(7) If Phil is traveling to London, he will bring his guitar.

Because *u*-theories predict unconditional presuppositions, from their perspective, the nature of the proviso problem flips. If the semantics directly generates the stronger presuppositions, what explains the weaker, conditional presuppositions observed in cases like (5-a)? The standard solution invokes local accommodation, which globally ‘cancels’ the presupposition. When the presupposition is locally accommodated within the consequent, the sentence as a whole no longer presupposes that Phil has a guitar. Instead, it yields a reading where if Phil is a musician, he has a guitar and brings it. Plausibility considerations here override the general preference for global accommodation: it is unlikely that the speaker would know about someone’s guitar ownership, while being ignorant about whether they are a musician.²

Although this solution accounts for why presuppositional conditionals do not always project unconditional presuppositions, a DRT-style *u*-theory faces significant challenges. One of these is its failure to generate genuine conditional presuppositions, which are empirically attested in these environments (Beaver 2001; see also Schlenker 2011 for related issues). As a result, the proviso problem remains an unresolved challenge for the *u*-view.

1.1 Present work

Both *c*-theories and *u*-theories can describe the core data in their own theoretical vocabulary, and both acknowledge that plausibility considerations play a role in what complex presuppositional sentences are ultimately felt to presuppose. Much of the research has therefore been focused on teasing out a real divergence in empirical predictions (see e.g., Schlenker 2011; Mandelkern 2016). We will bring a different type of data — data from processing — to bear on the issue.

In particular, we build on the fact that plausibility considerations aside, the two types of theories make different predictions about what a listener needs to accommodate with different kinds of conditionals (and other complex constructions). If a presupposition *p* is triggered in the left argu-

²An alternative solution, pursued by Geurts 1999, to assume that the presupposition is bound to an antecedent derived via a bridging inference (e.g., that being a musician entails having a guitar). Because bridging is not obviously extendable to cases we will study here, we put this option aside and focus on cancellation.

ment of a conditional, both camps expect listeners to accommodate p as is. If a presupposition p is triggered in the right argument, u -theories expect listeners to accommodate p , whereas c -theories predict that they accommodate something weaker, something conditional on the antecedent: ‘If A , then p ’. These predictions are summarized in Table 1.

Table 1: Predicted semantic presuppositions under the two views

	c -theories	u -theories
$If A_p, B$	p	p
$If A, B_p$	$if A, p$	p

The two theories thus differ on the strength of the proposition that needs to be accommodated when presuppositions are triggered in the right argument. The empirical challenge, however, lies in finding a way to measure this quantitatively. Here we approach this issue by using indicators of presupposition cancellation to infer the strength of projected presuppositions. Both theories agree that language users have in their repertoire a mechanism to get rid of or cancel ‘problematic’ presuppositions. This mechanism may involve local accommodation (e.g., Heim 1983) or the insertion of an ‘A-operator’ that converts presuppositions into assertions (e.g., Beaver & Krahmer (2001)). Here we use the neutral term ‘cancellation’, without commitment to the mechanism by which canceled readings are obtained.

Crucially, c -theories and u -theories not only predict different projection patterns from binary connectives, they also make different predictions about when such sentences require cancellation. Suppose a presuppositional conditional of the form ‘If A, B_p ’ is uttered in a context that is incompatible with p . U -theories would say that this is a situation that calls for cancellation. On c -theories, on the other hand, p being incompatible with the utterance context is not a problem so long as the weaker, conditional presupposition is compatible with the context.

The main reason presupposition cancellation is more straightforward to detect than the presupposed propositions themselves is that a substantive body of experimental work has shown that cancellation incurs a processing cost (Chemla & Bott 2013; Romoli & Schwarz 2015; Zehr & Schwarz 2016). A consistent finding is that the speed of processing presuppositional sentences is slower on cancelled readings than ones in which the presuppositions project globally. In Chemla & Bott’s (2013) experiment, for instance, participants had to evaluate whether a sentence like (8) is true. (8) involves a negated factive verb (*realize*) embedding a proposition that common knowledge tells us is false (e.g., that elephants are birds). If this factive presupposition is accommodated globally, as in (8-a), the sentence should be judged as false. If on the other hand the presupposition is cancelled (via local accommodation or A-operator insertion), the sentence has the reading in (8-b), which can be judged true. Crucially, responding “true” in these cases took significantly longer than responding “false”, a delay the authors attribute to the cost of cancellation in (8-b).

- (8) Zoologists do not realize that elephants are birds.
- a. [Elephants are birds] and NOT [zoologists believe so] (false)
 - b. NOT [(Elephants are birds) AND (zoologists believe so)] (true)

With cancellation, then, we have a way of detecting effort, even when that effort is not introspectively perceived. If presuppositions triggered in either argument of a binary connective project

unconditionally, as predicted by *u*-theories, whenever that unconditional presupposition is incompatible with the context, we expect a “cancellation cost” during processing. If only presuppositions triggered in the left argument project unconditionally, as in *c*-theories, we expect an asymmetry in processing signature: contexts incompatible with the unconditional presupposition should result in a cancellation cost only when it originates in the first argument.

We investigate whether these differences posited by *c*-theories are detectable in processing of presuppositional conditionals and disjunctions. In our study, we present participants with complex sentences with a presupposition *p* triggered in either the left or right argument of the connective, in contexts that are incompatible with *p* (but compatible with a weaker, conditional proposition). We use participants’ responses and response times as an indicator of when cancellation was necessitated, and in turn, when the complex sentence on the whole presupposes *p*. The first two experiments, reported in Section 2, investigate the processing of presuppositional conditionals.³ In Section 3, we extend the investigation to disjunctions. Finally, Section 4 discusses the implications of the present work.

2 Processing presupposition in *if*-conditionals

We first examine participants’ processing of presuppositional conditionals, comparing their behavior when a presupposition is triggered in the left (antecedent) vs. right (consequent) arguments. We ask how participants respond to such sentences when the context is globally incompatible with the unconditional presupposition.

2.1 Experiment 1

2.1.1 Participants

120 native English speakers participated in this study. Participants were recruited via Prolific and received \$5.00 for per 30 minutes of participation. An additional 15 participants were tested but excluded for failing to pass attention checks (<70% accuracy with filler items, for which “No” is the expected response).

2.1.2 Methods

Workflow We constructed an Outcome Evaluation Task, where participants read a set of sentences and had to evaluate an outcome image on a subsequent display relative to those sentences. The task was set up as a dress-up game with parameters on how to dress up a stick-figure. Participants first saw the (not-yet-dressed) stick figure and read three instruction sentences on how it should be dressed up (Figure 1).

³In this paper, we follow the literature on presupposition projection in assuming a material analysis of conditionals. We acknowledge, however, that a more precise analysis of conditionals might be as restrictors of (overt or covert) modals, in line with Kratzer (1986). Adopting such an analysis would turn this into a question not about projection from the two arguments of binary connectives, but about projection from the restrictor and nuclear scope of (universal) quantifiers. *c*-theories predict an asymmetry in this case, as well, with presuppositions projecting more weakly from the scope compared to the restrictor. It is less clear what the predictions of *u*-theories are here; see discussion in Beaver (2001).

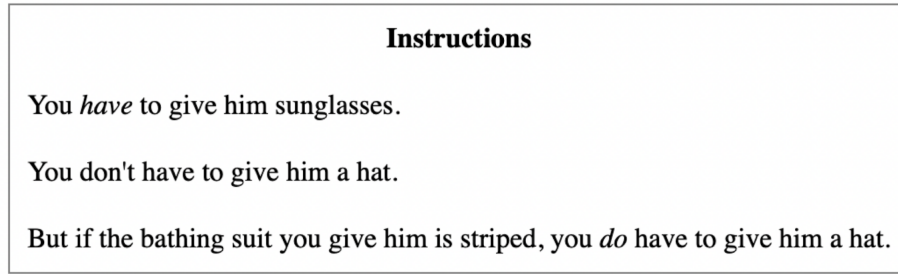


Figure 1: Workflow in Experiment 1 with a sample first scene

After 3 seconds, a button appeared below the instructions that let participants advance to the next page. They were, however, allowed to inspect the instruction sentences for as long as they wanted.⁴ The second page, which presented the outcome of the dress-up, depicted a stick figure wearing some set of clothing items. The instructions remained on screen, above the image, but in gray text with smaller font size (Figure 2). On the same page, participants were asked to decide, as quickly as they can, whether the dress-up was in accordance with the instructions, by selecting “Yes” or “No”. They had 6 seconds to make a selection.

Materials, design and procedure The nature of the instruction sentences and the nature of the outcome were varied across trials. As mentioned, there were three sets of parameters. The first sentence, which had a consistent form “You have to give him *Item*₁”, was always met in the outcome. The second was met some of the time, and consistently had the form: “You don’t have to give him *Item*₂”. The main role of this sentence was to support the uncertainty inference of the third sentence, which was a conditional featuring a new item (*Item*₃) as well as *Item*₂ from the second sentence.

This third instruction sentence crucially figured into our manipulations. In our critical trials, the conditional featured a definite description, the presupposition trigger, in either the left argument (9-a) or the right (9-b). Environment (left vs. right argument) was a between-subjects factor.

- (9) a. If the bathing suit you give him is striped, you have to give him a hat.
b. If you give him a hat, the bathing suit you give him has to be striped.

There were furthermore three Trial Types (p_true; p_critical and control) that varied in the degree to which the outcome image matched the third instruction sentence. In the p_true Trial Type, this sentence was a presuppositional conditional, and all items from this sentence appear in the dress-up outcome in a way that (i) supports the unconditional presupposition of the conditional and (ii) renders the entire conditional true. In p_critical Trials, the instruction sentence is identical to that in p_true, i.e. a presuppositional conditional, but neither item mentioned in the sentence appears in the dress-up outcome. This means that the unconditional presupposition is not supported, though a weaker, conditional presupposition is. Finally, control trials feature a non-presuppositional con-

⁴Text selection was disabled on the webpage, so participants would not be able to copy and paste the sentences to a local text editor.

Instructions

You *have* to give him sunglasses.

You don't have to give him a hat.

But if the bathing suit you give him is striped, you *do* have to give him a hat.

This is how someone dressed the stickman.



Did they do an OK job following instructions?

Yes

No

Figure 2: Workflow in Experiment 1 with a sample second scene

ditional with same mentioned items ($Item_3$ is presented with an indefinite) and outcome image to p_critical. See Table 2.

We created a list of 27 articles of clothing and accessories and randomly selected subsets of 3 to create 72 test items, 12 trials per condition. This was meant to minimize any plausibility relation between the two arguments. Additionally, we included false fillers also involving conditionals, but with “No” as the expected response; these were used as attention-checks and figured into identifying and excluding inattentive participants. See Appendix for full description of the fillers.

The experiment began with 4 practice trials in fixed order with feedback. The order of all experimental items was randomized. Participants received feedback after half of the filler items.

Analysis. Data were analyzed using R, version 3.6.3 (R Core Team, 2021). Mixed effects modeling was done using the *lme4* package (version 1.1–28; Bates, Mächler, Bolker, and Walker 2014).

Table 2: Sample materials for Experiment 1

Consistent across Trial Type	You <i>have</i> to give him sunglasses. You don't have to give him a hat...		
	Outcome Image	Environment (Argument of Connective)	
		Antecedent (left)	Consequent (right)
p_true		But if <u>the bathing suit you give him</u> is striped, you do have to give him a hat.	But if you do give him a hat, <u>the bathing suit you give him</u> has to be striped.
p_critical		But if <u>the bathing suit you give him</u> is striped, you do have to give him a hat.	But if you do give him a hat, <u>the bathing suit you give him</u> has to be striped.
control		But if you give him a bathing suit, you do have to give him a hat.	But if you do give him a hat, you have to give him a bathing suit.

We fit a logistic regression model on Yes-responses, which included fixed effects of Environment, Trial Type and their interactions. Trial Type and Environment were simple coded with p_critical and Consequent, respectively, as reference levels. We analyzed RT data using a generalized mixed-effects model with a log-distribution, predicting (raw) Yes-response times (RTs) as a function of the same fixed effect structure as the Yes-response model. Both models included random intercepts for Participant and Item; this was the maximal random effect structure that converged.

2.1.3 Predictions

We collect and analyze two dependent measures: judgments of instruction compliance (Yes/No) and response times.

When outcomes adhered to instructions and presuppositions (e.g., p_true trials), both theories predict high rates of “Yes” responses across environment. Similarly, in control trials, which had false antecedents that rendered the conditionals on the whole true, “Yes” responses are predicted. With p_critical trials, the expectations vary. Across environments, the outcome in these trials failed to *globally* support the presupposition of the sentence. On both theories, if the presupposition was triggered in the antecedent, it becomes a presupposition of the conditional on the whole. It is possible here for participants to respond “No” on the grounds that an item the instruction-giver took for granted was missing. Alternatively, they can respond “Yes”, but only if they access a canceled reading. If, e.g., the presupposition is locally accommodated within the antecedent, the instruction sentence in (10-a) would have the interpretation in (10-b). This would make such cases truth-conditionally comparable to the control trials: the antecedent would be false at outcome, the conditional on the whole would be true (no bathing suit, no hat).

- (10) a. If the bathing suit you give him is striped, you do have to give him a hat.
 b. If there is a bathing suit you give him and it is striped, you do have to give him a hat.

For presuppositions triggered in the consequent, theoretical predictions diverge. On *c*-theories, if the antecedent is false, the sentence on the whole presupposes nothing. Thus, an outcome where neither a bathing suit nor a hat is given is one where the presuppositions of the instruction sentence are already met, and we simply expect a “Yes” response. By contrast, *u*-theories predict an unconditional presupposition. A “Yes-response”, requires cancellation, yielding an interpretation like (11-b) for a sentence like (11-a).

- (11) a. If you do give him a hat, the bathing suit you give him has to be striped.
 b. If you do give him a hat, there is a bathing suit you give him and it has to be striped.

These predictions are summarized in Table 3. Critically, an asymmetry in “Yes” responses, with lower “Yes”-rates when the trigger is in the left argument, is straightforwardly predicted only on *c*-theories.

Table 3: Predictions for p_critical trials across theories



	p in Left argument	p in Right argument
	If <u>the bathing suit you give him</u> is striped, you do have to give him a hat.	If you do give him a hat, <u>the bathing suit you give him</u> has to be striped.
<i>c</i>-theories	p = there is a bathing suit; context-inconsistent → cancellation needed for Yes response	p = if there is a hat, then there is a bathing suit; context-consistent → Yes response
<i>u</i>-theories	p = there is a bathing suit; context-inconsistent → cancellation needed for Yes response	p = there is a bathing suit; context-inconsistent → cancellation needed for Yes response

At the same time, both theories can account for “Yes” responses in both environments, which limits the informativeness of response rates alone. But because the theories differ in *when* a “Yes” response requires cancellation, response times could offer additional insight. Specifically, *c*-theories predict an environment-based difference in response times for p_critical trials (antecedent

> consequent), as only the antecedent environment gere demands cancellation. Thus, even if “Yes” response rates are identical across conditions, response times could reveal differences in the processes underlying participants’ judgments.

2.1.4 Results

For all three target trial types, across environments, participants overwhelmingly responded “Yes”, with no statistically significant differences across conditions. See Table 4. Notably, the p_critical condition, the only target trial type where a “No” response was in principle justifiable, had high Yes-response rates. This suggests that participants were charitable: outcomes without the item mentioned in the presupposition trigger were treated as instruction-adherent.

Table 4: Mean rates [and 95% CIs] of responding “Yes” to target items; Experiment 1

	Antecedent	Consequent
p_true	97.7% [96.6%-98.8%]	96.6% [95.3%-97.9%]
p_critical	97.4% [96.3%-98.6%]	97.7% [96.6%-98.8%]
control	97.2% [96.0%-98.4%]	97.9% [96.8%-99.0%]

Our second measure was the time it takes for participants to respond “Yes” across conditions. Response times for are summarized in Table 5. RTs for p_critical trials are represented by the blue lines in Fig. 3.

Table 5: RTs for Yes responses (ms) from Experiment 1

	Antecedent	Consequent	Ant-Cons Difference
p_true	2527.84	2516.19	11.65
p_critical	2239.91	2115.00	124.91
control	2167.84	2049.39	118.45

Table 6 presents the fixed effects results for our regression modeling of the RT data. The model found two significant effects. First, participants were slower to say “Yes” to p_true trials compared to p_critical trials. Second, Environment interacted with Trial Type for the contrast between p_true and p_critical: whereas there were no environment-driven differences in p_true trials, in p_critical trials participants were slower to say “Yes” in the Antecedent environment compared to the Consequent. We did not, however, find a parallel interaction when comparing p_critical and control. In control trials, like p_critical trials, the time taken to arrive at a Yes-response in the Antecedent environment was longer than the Consequent.

2.1.5 Discussion

In Experiment 1 critical trials, participants tended to judge an outcome as instruction-adherent even when it did not support the presuppositions of the instruction sentence in an unconditional form. Though these judgments did not differ based on where the presupposition was triggered, we found an asymmetry in response times in line with what *c*-theories predict. Participants were faster when

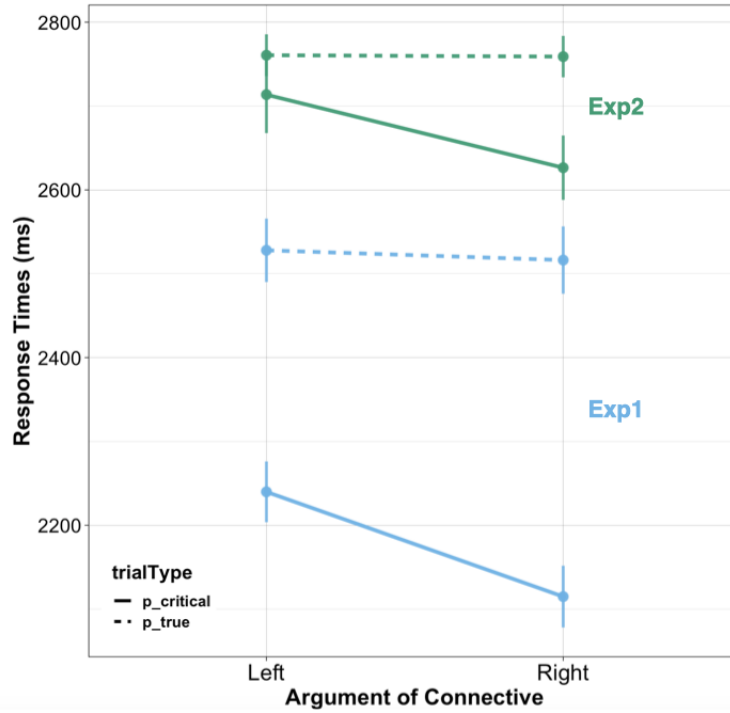


Figure 3: Mean RTs by argument of the connective across Experiments 1 and 2. Solid lines represent p_critical trials and dashed lines represent p_true trials. Error bars represent +/- 1 standard error.

Table 6: Parameter values for fixed effects in mixed-effects regression model of RTs in Experiment 1

	Estimate	Std. Error	<i>t</i>	<i>p</i>
Intercept	7.61	0.05	146.89	<.001
Environment	0.07	0.06	1.08	0.28
p_critical vs. control	-0.03	0.03	-0.85	0.40
p_critical vs. p_true	0.20	0.03	6.07	<.001
Environment * p_critical vs. control	-0.01	0.02	-0.26	0.80
Environment * p_critical vs. p_true	-0.07	0.02	-3.18	.001

the trigger was in the right argument (consequent) than the left. This environment-based difference can be explained if participants did not need to cancel presuppositions when they were triggered in the right argument, but they did need to do so with the left argument. The response time latencies could be seen as a cancellation cost, in line with what prior work.

Consistent with this interpretation, participants did not differentiate between the two arguments — in judgments or response times — when the outcome supported the presupposition. However, in the control trials, which had no presupposition triggers, participants did differentiate between the environments. Recall that the control trials involved non-presuppositional conditionals with

both a false antecedent and a false consequent. The falsity of the antecedent rendered the entire conditional true, and the expected response was "Yes." Crucially, these trials were functionally identical across environments, so any environment-based asymmetry is unexpected. The presence of such an asymmetry suggests that third factors could be involved, potentially undermining our interpretation of the asymmetry in the presuppositional trials.

One alternative explanation is that participants relied on a shallow evaluation strategy, responding primarily based on items mentioned in the antecedent. When they noticed that a (non-presupposed) item mentioned in the antecedent did not appear in the outcome (i.e., in all trials in the p_critical_consequent condition and the control condition), they immediately responded "Yes" using their logical knowledge that a false antecedent results in a true conditional. If this was the case, we might re-characterize the RT advantage in those trials as a False Antecedent (FA) advantage. This would mean that the response times for p_critical_consequent trials do not provide information about the processing of presuppositions, as participants may not have carefully read or processed the presupposition trigger in the first place. Experiment 2 addresses this concern with a revised design.

2.2 Experiment 2

Experiment 2 used the same task and target materials as Experiment 1. But to reduce the possibility of the FA-advantage discussed above, we opted for a block design. Participants saw a long sequence of trials before encountering a trial with a false antecedent. We also included filler trials involving other types of complex sentences to preclude form-based strategies. The reduced frequency of seeing trials involving a false antecedent and the longer interval between each such trial means that an FA-strategy will be less useful (assuming that such a strategy develops over time rather than immediately).

Furthermore, the new design could allow for a more robust detection of the application of cancellation mechanisms. Recent work has suggested that frequent exposure to trials requiring cancellation primes its invocation in later trials, even when the operation is unnecessary for felicity (Zehr & Schwarz 2016). In other words, over the course of an experiment, cancellation could become a default rather than the last-resort repair mechanism it is thought to be. If this is correct, participants in our task could assume default "presupposition-less" readings after having had to access such readings enough times. If participants choose to cancel presuppositions across the board by default, the sought-after asymmetry might not be detected. The revised design also aimed at reducing such anticipatory applications of cancellation mechanisms.

2.2.1 Participants

We recruited 200 participants via Prolific for this experiment. Sample size was determined based on the number of observations for critical items. In Experiment 1, 120 participants completed 12 p_critical trials each, resulting in 1,440 observations. In Experiment 2, with 6 p_critical trials per participant, 200 participants yielded 1,200 observations, bringing us close to the number from Experiment 1. Four additional participants were excluded for having less than 70% accuracy on fillers.

2.2.2 Methods

Materials, design and procedure Experiment 2 used the same task and target trial types as Experiment 1: p_true, p_critical and control. There were four main differences. First, environment was a within-subjects factor, with each participant seeing both antecedent and consequent trials. Second, we systematically reduced the number of trials involving a false antecedent. Each participant saw 3 each of p_critical_antecedent, p_critical_consequent and control trials.⁵ Third, in addition to the false conditional fillers from Experiment 1, we included fillers involving conjunctions. In these trials, the outcome varied based on whether items described in one or both conjuncts were included, leading to responses of either "Yes" or "No." Finally, these trials were presented in blocks of 7 trials each, in a pre-determined order schematized in Table 7.

Table 7: Trial order in the first 3 blocks in one of the two lists in Experiment 2

	Sequence of 6 trials	Trial with a false antecedent
Block 1	(p_true, false conditional filler, conjunction filler) ×2	p_critical: trigger in Antecedent
Block 2	(p_true, false conditional filler, conjunction filler) ×2	control: no presupposition triggers
Block 3	(p_true, false conditional filler, conjunction filler) ×2	p_critical: trigger in Consequent

The first 6 trials in each block were a combination of different types of trials with a true antecedent, and the final trial was either p_critical_antecedent, p_critical_consequent, or control. We counterbalanced whether the first p_critical trial seen had the trigger in the left vs. right argument across 2 lists. The order of the other trials within each block was pseudo-randomized. There were 9 blocks, with a total of 63 trials across them. The total number of Yes/No expected responses over the task was balanced. Block order and trial order within block remained the same for all participants within a list.

The experiment began with 4 practice trials, for which explicit feedback was provided to the participants regardless of whether their response was correct. Participants needed to answer all training trials correctly in order to proceed.

Analysis Because the design was not fully crossed, we planned two separate analyses each for our two dependent variables (DVs), yielding four models in total. For each DV (Yes-Response, RTs), we first compared p_critical vs. p_true across environments by analyzing the subset of data including only these two trial types. The two factors were simple-coded, with Antecedent as reference level for Environment p_critical the reference level for Trial Type. The second model compared the three false antecedent trials: p_critical_antecedent, p_critical_consequent, and control. These models included FA-Trial Type as a fixed effect with three levels, simple-coded with p_critical as the reference level. Models included participant random intercepts and random slopes of Trial Type and Environment.⁶

⁵Note that in Experiment 2, we do not differentiate control trials by environment. The only reason for splitting control trials by environment in Experiment 1 was its between-subjects design.

⁶Item was fully correlated with condition in this design, so we do not include item random effects.

2.2.3 Results and discussion

As in Experiment 1, participants overwhelmingly responded “Yes”. However, p_critical trials yielded lower Yes-responses than p_true trials and importantly, within p_critical trials, the Antecedent environment yielded somewhat lower Yes-responses than Consequent; see Table 8.

Table 8: Mean rates [and 95% CIs] of responding “Yes” to target items; Experiment 2

	Antecedent	Consequent
p_true	97.9 [97.2%-98.6%]	98.1% [97.4%-98.7%]
p_critical	94.7% [92.9%-96.6%]	96.1% [94.5%-97.7%]
control	97.3 [96.0%-98.6%]	

This difference was statistically significant. The comparison between p_true and p_critical trials across environments revealed an interaction of Environment and Trial Type, driven by the fact that Yes-responses were asymmetrically lower for p_critical trials in the Antecedent environment. See Table 9 for full model output.

Table 9: Parameter values for fixed effects for p_true vs. p_critical comparison of Yes-Responses in Experiment 2

	Estimate	Std. Error	<i>z</i>	<i>p</i>
Intercept	2.98	0.29	10.25	<.001
Environment	4.40	1.17	3.78	< .001
Trial Type	1.25	0.51	2.42	0.02
Environment * Condition	-4.12	1.34	-3.06	0.002

With RTs, we saw similar numerical trends as in Experiment 1 (see Table 10) and Figure 3; green. Responding “Yes” to p_critical trials in the Antecedent environment took 87.12 ms longer than the Consequent environment in the p_critical condition, whereas this difference is only 1.68 ms in the p_true condition. This difference was not statistically significant, however. The model comparing p_true and p_critical trials across environments, summarized in Table 11 did not reveal any significant main effects or interaction between Environment and trial Type.

Table 10: RTs for Yes responses (ms) from Experiment 2

	Antecedent	Consequent	Ant-Cons Difference
p_true	2760.58	2758.90	1.68
p_critical	2713.58	2626.46	87.12
control	2420.58		NA

The second model, which compared across the three false antecedent trials, revealed a significant difference between p_critical.consequent and control. Participants were faster to respond “Yes” to control trials compared to p_critical consequent, suggesting that they were not relying on the interpretive strategies for the two trial types. See Table 12 for full model output.

Table 11: Parameter values for fixed effects for p_true vs. p_critical comparison of RTs in Experiment 2

	Estimate	Std. Error	<i>t</i>	<i>p</i>
Intercept	7.90	0.02	319.00	<.001
Environment	0.002	0.02	0.17	0.87
Trial Type	-0.02	0.02	-0.98	0.33
Environment * Condition	-0.03	0.02	-1.36	0.18

Table 12: Parameter values for fixed effects from FA-trial type comparison of RTs, Experiment 2

	Estimate	Std. Error	<i>t</i>	<i>p</i>
Intercept	7.83	0.03	286.11	<.001
p_critical_cons vs. p_critical_ant	0.02	0.03	0.88	0.38
p_critical_cons vs. control	-0.09	0.02	-4.41	<.001

In this experiment, unlike Experiment 1, we found an asymmetry across the two arguments in Yes-response rates, but not in response times. When the outcome failed to verify a presupposition *p*, participants were more likely to say “No” if *p* was triggered in the Antecedent compared to the Consequent. Furthermore, participants in this experiment could not have employed a false antecedent strategy: we observed a significant difference in response times between non-presuppositional control trials and critical trials with a presuppositional consequent, with the latter resulting in longer response times. This indicates that participants continued processing the presupposition trigger instead of stopping after encountering the false antecedent. At the same time, we did not find a statistically significant difference in response times between the left and right arguments, despite the numerical trends pointing in the right direction. In the next section, we discuss reasons for the two critical between-experiment differences.

2.3 Discussion: projection in conditionals

In two experiments, we explored cancellation signatures to identify how presuppositions project from the left and right arguments of conditionals. Participants read conditionals such as (12), with presuppositions triggered either in the antecedent (12-a) or the consequent (12-b), presented as dressing instructions. They were then shown dress-up outcomes that did not globally support the presupposition. We asked two questions: (i) whether participants judged the outcome as instruction-compliant differentially based on whether the presupposition triggered in the antecedent compared to the consequent; and (ii) when they judged the outcome as instruction compliant, whether it took them longer to respond with presuppositions triggered in the antecedent compared to the consequent. Both these asymmetries would be more in line with *c*-theories, which predict that in these scenarios, cancellation is *selectively* required for presuppositions triggered in the antecedent.

- (12) a. If the bathing suit you give him is striped, you have to give him a hat.
 b. If you give him a hat, the bathing suit you give him has to be striped.

Our two experiments yielded mixed evidence that we first summarize and then interpret. Regarding

explicit outcome evaluations, upon seeing sentences like (12-a) and (12-b), followed by an outcome lacking a bathing suit, participants almost always judged it as complying with the instructions. In Experiment 1, this judgment was unaffected by the location of the presupposition trigger. However, in Experiment 2, participants were less likely to consider the outcome instruction-compliant when the instruction had the form in (12-a) (antecedent environment) compared to (12-b) (consequent environment). For reaction time data on these “Yes” responses, Experiment 1 revealed a clear effect: participants were slower to respond “Yes” when the presupposition was triggered in the antecedent. In Experiment 2, this response time effect did not emerge, though the data showed a numerical trend in the same direction. Across both experiments, we found no asymmetry between antecedent and consequent environments when the presupposition was supported in the outcome — i.e. when the outcome included a bathing suit.

Our dependent measures showed a complementary distribution across the two experiments. In each experiment, we observed an asymmetry in one of our two measures — rates of responding “Yes” and response times to say “Yes” — but not both. The fact that each experiment reveals one type of effect makes it difficult to dismiss either finding as purely artifactual. For example, even if the response time effect in Experiment 1 were attributable to an FA-strategy — explaining its absence in Experiment 2 — we would still need to account for the judgment asymmetries that show up in Experiment 2.

How, then, can we make sense of the overall pattern? A key consideration is the statistical relationship between our two dependent measures. We found a significant response time effect only in Experiment 1 where participants responded “Yes” across environments at equally high rates. In Experiment 2, participants responded “Yes” less often in the antecedent environment. This reduction in “Yes” responses meant fewer measurements for the reaction time analysis compared to Experiment 1. Additionally, Experiment 2 started with fewer p_critical trials overall. Combined with the lower “Yes” rates, this resulted in approximately 18% fewer observations for the response time analysis, raising the possibility that Experiment 2 may have been underpowered to detect the same response time effect observed in Experiment 1.

A second consideration is the theoretical relationship between the judgments and the response time signature of cancellation under *c*-theories versus *u*-theories. According to *c*-theories, participants who responded “no” in antecedent trials may have done so because canceling the presupposition and responding charitably was too costly. In other words, the trials where one would expect the largest cancellation cost in terms of response times might also have been the trials where participants opted out by responding “No.” This increased tendency to respond “No” in Experiment 2 could therefore be further reason why a reaction time effect was not observed in that experiment.

While we find it more challenging to reconcile the full pattern of data with *u*-theories, we believe it is still possible. This is primarily because the semantic contributions of the antecedent and consequent in conditionals are inherently different, not only in their projection patterns but also in how they contribute to the conditional’s overall meaning (e.g., their distinct entailment patterns). These differences make alternative explanations plausible, where an observed difference in a response variable might arise from unrelated factors.

To reduce the plausibility of such alternatives, we need to shift to a linguistic context that where, modulo presupposition projection asymmetries, the semantic properties of the two arguments can be taken as more comparable. In Experiment 3, we turn to another binary connective — disjunction — where the predictions of *c*-theories and *u*-theories for projection similarly diverge. Beyond

extending to a new environment, this will also serve as a conceptual replication of our current findings.

3 From conditionals to disjunctions

In this section, we extend our investigation of presupposition projection to another binary connective, disjunction. Like conditionals, disjunctions are another context in which proviso effects arise. Often, presuppositions triggered in the second disjunct appear to project more weakly than those in the first disjunct, as illustrated in (13). When a presupposition is triggered in the first disjunct of a disjunction (A or B), it becomes the presupposition of the entire disjunction, as seen in (13-a). However, if the same trigger appears in the second disjunct, it projects conditionally, in the form ‘if not A, then B’ (13-b).

- (13) a. Either Phil brings his guitar on the trip or he isn’t a real musician.
Presupposes: Phil has a guitar
 b. Either Phil is not a musician or he will bring his guitar on the trip.
Presupposes: If Phil is a musician, he has a guitar

This conditional presupposition is what is predicted on *c*-theories, but the prediction is complicated by examples like (14) where a presupposition triggered in the 2nd disjunct projects unconditionally. This, without further elaboration, seems more in line with *u*-theories.⁷

- (14) Either Phil is not traveling to London, or he will bring his guitar on the trip.
Presupposes: Phil has a guitar

Following the logic of our experiments so far, we also predict differences in recruitment of cancellation mechanisms across the two theoretical camps. A context that doesn’t support the unconditional presupposition *p* should selectively trigger cancellation when *p* is part of the first disjunct.

3.1 Experiment 3

3.1.1 Participants

150 participants, recruited via Prolific, took part in this experiment. The number of p_critical items was increased to 8 in Experiment 3, allowing us have the same number of observations as Experiment 2 (i.e., ×1200) with fewer participants.

⁷It should be noted here that some *c*-theories derive conditional semantic presuppositions also for the first disjunct (Schlenker 2008, Fox 2008, Rothschild 2008, Chemla 2009). On these theories, any left-right asymmetry between the two arguments results as a side-effect of incremental processing. Because presuppositions are checked incrementally, as soon as the trigger is encountered, presuppositions triggered in the left argument of a disjunction ‘A_p or B’ end up getting evaluated relative to the global context (as people parse this portion of the sentence prior to parsing the second disjunct), though semantically such sentences are associated with the presupposition ‘if ¬B, *p*’. We return to these theories in the General Discussion.

3.1.2 Methods

Workflow To make disjunctive sentences more natural, this variant of the Outcome Evaluation Task involved evaluating whether the outcome matched a set of predictions made about the dress-up. Participants first saw the (not-yet-dressed) stick figure and read two sentences that represented guesses about how it will be dressed. They then saw the outcome display and had to indicate whether the predictions were born out in the outcome. Other aspects of the workflow were identical to Experiments 1 and 2.

Materials, design and procedure Participants saw two prediction sentences on each trial. The first was a simple sentence of the form “He wears *Item₁*”, where the mentioned item was always part of the outcome. The second and crucial sentence was a disjunction, featuring two additional items, one in each disjunct. In the critical trials, one of these items was presented in a definite description, which served as the presupposition trigger. We manipulated within-subjects Environment in which trigger appeared (1st Disjunct vs. 2nd Disjunct) and Trial Type (p_true, p_critical, control). In p_true trials, the item described by the presupposition trigger appears in the outcome in a manner that supports the unconditional presupposition and renders the disjunction true (the item mentioned in the other disjunct does not appear in the outcome). The opposite pattern is displayed by the p_critical trials: the item described by the presupposition trigger does not appear in the outcome. The unconditional presupposition is thus not supported. The (non-presupposed) item mentioned in the other disjunct is part of the outcome, thus verifying the disjunction. Additionally, we included non-presuppositional control trials, where the disjunction on the whole is verified by the same disjunct as p_critical. The conditions are schematized in Table 13.

Table 13: Sample materials for Experiment 3

Consistent across Trial Type	He has an umbrella.		
Trial Type	Outcome Image	Environment (Argument of Connective)	
		1 st Disjunct (left)	2 nd Disjunct (right)
p_true		<u>The socks he wears</u> are red, or he wears a scarf.	He wears a scarf, or <u>the socks he wears</u> are red.
p_critical		<u>The gloves he wears</u> are blue, or he wears socks.	He wears socks, or <u>the gloves he wears</u> are blue.
control		He wears blue gloves, or he wears socks.	He wears socks, or he wears blue gloves.

All participants saw 4 trials per condition. In addition to the critical trials, we included 4 types of filler trials, described in more detail in the Appendix. Trials were presented in blocks, similarly

to Experiment 2. Each block contained 7 trials. The first 6 was a mix of fillers, *p*_true and control trials. The 7th trial in each block was a *p*_critical trial in one of the two environments (Table 14). There were a total of 56 trials across 8 blocks, with balanced Yes/No expected responses across the experiment. Across two lists, we counterbalanced whether the first *p*_critical trial encountered had the presupposition trigger in the 1st or 2nd Disjunct a two lists. Otherwise, block order and order of non-critical trials within each block were the same for all participants.

Table 14: Trial order in the first 2 blocks in one of two lists in Experiment 2

	Sequence of 6 trials	<i>p</i> _critical trial
Block 1	<i>p</i> _true, filler (x3) control, filler	<i>p</i> _critical, trigger in 1st Disjunct
Block 2	<i>p</i> _true, filler (x3) control, filler	<i>p</i> _critical, trigger in 2nd Disjunct

The experiment began with 7 practice trials with feedback. Two of these trials involved disjunctions where both disjuncts were true, highlighting that the inclusive reading of *or* is acceptable. Participants had to answer all training trials correctly in order to proceed to the experimental phase.

Analysis Yes-responses were analyzed via a mixed-effects logistic regression, with Environment, Trial Type and their interactions as fixed effects. Both factors were simple-coded, with 1st Disjunct as reference level for Environment and *p*_critical the reference level for Trial Type. RTs were analyzed using a mixed-effects regression with a log-distribution. The same fixed effects and contrast-coding scheme was used here as the model on Yes-responses. Both models included random intercepts and slopes for participants.

3.1.3 Results

Because predictions about presupposition projection are based on a basic, inclusive reading of *or*, we excluded data from 5 participants who responded “No” to filler items involving two true disjuncts. These participants seemed to be rejecting the inclusive reading of *or* despite explicit training.⁸

Yes-rates for the target conditions are summarized in Table 15. Table 16 summarizes the results from a mixed-effects logistic regression model fit on these data. The model revealed two significant effects. Participants responded “Yes” at higher rates on control trials compared to *p*_critical. There was also a significant interaction of environment and the contrast between *p*_true and *p*_critical trial types. This effect was due to an environment-based difference in only the *p*_true trials, where there were more Yes-responses in the 1st Disjunct compared to the 2nd Disjunct.

Response time results are summarized in Table 17 and Figure 4. Table 18 presents the results of our regression modeling of these data. We found a significant interaction of Environment and the contrast between *p*_critical and *p*_true: participants were faster to respond “Yes” when the presupposition trigger was in the 2nd Disjunct compared to the 1st in the *p*_critical trials, but the pattern was the opposite for *p*_true trials. We also found an interaction of Environment and the contrast between *p*_critical and control. For *p*_critical trials, the 2nd Disjunct environment yielded faster responses; no such asymmetry was found with the control trials.

⁸Corroborating this interpretation, a subset of them left post-experiment comments saying that they believed the inclusive reading from the training is “actually grammatically incorrect”.

Table 15: Mean rates [and 95% CIs] of responding “Yes” to target items; Experiment 3

	1st Disjunct	2nd Disjunct
p_true	97.9% [96.7%-99.1%]	95.3% [93.6%-97.1%]
p_critical	89.9% [87.5%-92.4%]	89.6% [87.1%-92.2%]
control	97.7% [96.5%-99.0%]	98.6% [97.6%-99.6%]

Table 16: Parameter values for fixed effects in mixed-effects regression model of Yes-rates, Experiment 3

	Estimate	Std. Error	<i>z</i>	<i>p</i>
Intercept	2.61	0.22	12.04	<.001
Environment	0.13	0.30	0.42	0.67
p_critical vs. control	2.22	0.39	5.67	<.001
p_critical vs. p_true	0.90	0.25	3.55	<.001
Environment * p_critical vs. control	-0.49	0.51	-0.97	0.33
Environment * p_critical vs. p_true	0.91	0.42	2.15	0.03

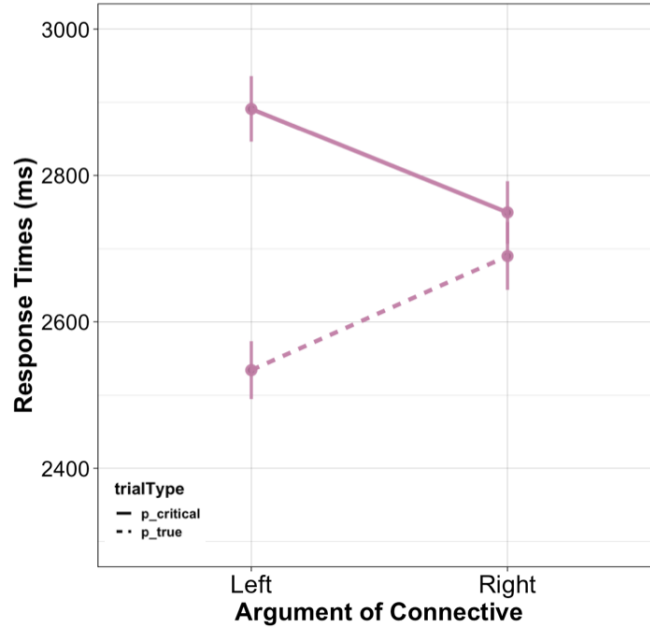


Figure 4: Mean RTs by argument of the connective in Experiment 3. Solid lines represent p_critical trials and dashed lines represent p_true trials. Error bars represent +/-1 standard error.

3.2 Discussion

Experiment 3 extended our investigation to disjunctions to examine the nature of presuppositions projected from the left versus right arguments of *or*. As in previous experiments, we focused on signatures of presupposition cancellation, exploring whether and when contexts incompatible

Table 17: RTs for Yes responses (ms), Experiment 3

	1st Disjunct	2nd Disjunct	1st-2nd Difference
p_true	2534.07	2690.02	-155.95
p_critical	2891.01	2749.51	141.50
control	2377.82	2450.82	-73.00

Table 18: Parameter values for fixed effects in mixed-effects regression model of RTs in Experiment 3

	Estimate	Std. Error	<i>t</i>	<i>p</i>
Intercept	7.90	0.03	273.50	<.001
Environment	0.05	0.02	2.89	0.004
p_critical vs. control	-.12	0.02	-5.64	<.001
p_critical vs. p_true	-0.03	0.02	-1.59	0.11
Environment * p_critical vs. control	-0.09	0.23	-3.85	<.001
Environment * p_critical vs. p_true	-0.11	0.02	-4.60	<.001

with an embedded presupposition lead to cancellation. While we did not find an environment-based asymmetry in “Yes” response rates, we found that participants responded more slowly when the outcome failed to verify an unconditional presupposition p and p was triggered in the first disjunct compared to the second. This difference mirrors the left-right asymmetry observed with conditionals in Experiment 1 — even with the blocked design as in Experiment 2, intended to minimize learning effects.

In addition to this predicted difference, we found an unexpected difference in contexts where the outcome verified the unconditional presupposition. Participants responded “Yes” faster and more often when the presupposition was triggered in the first disjunct compared to the second disjunct — exactly the opposite pattern observed in contexts where the presupposition was unmet. The 1st Disjunct p_true trials, which have the trigger in the first disjunct, also involve disjunctions verified by the first disjunct. That is, the clothing items mentioned in the first disjunct is what is present in the outcome and renders the entire disjunction true. This is a trait the 1st Disjunct p_true trials share with the p_critical 2nd Disjunct trials, where again, it is the items mentioned in the first disjunct that are present in the outcome. Could the observed differences across environments then simply be due to a 1st Disjunct advantage, meaning participants are faster whenever the outcome supports the first disjunct? Not straightforwardly, because in our control trials, we detect no difference between cases where the outcome supports the first disjunct (2ndDisjunct_control) and those where it supports the second disjunct (1stDisjunct_control).

The differences observed in p_true and p_critical trials must thus be tied to their presuppositionality. Though this was not a statistical comparison we made, looking at the two environments separately shows that there was a notable difference (357ms) between trials where a presupposition in the first disjunct is supported by the outcome (p_true_1stDisjunct) versus those where it is not (p_critical_1stDisjunct). But we see a much smaller difference (60ms) for supported versus unsupported presuppositions in the second disjunct (p_true_2ndDisjunct vs. p_critical_2ndDisjunct). Tentatively, we interpret this to mean that, in addition to the cancellation cost we are interested

in, there may also be a processing advantage for *verified* presuppositions. More concretely, starting with presuppositions that are contextually supported might facilitate processing, though this hypothesis requires further investigation.

4 General Discussion

4.1 Summary of findings and implications

In this study, we used cancellation signatures as a tool to investigate presupposition projection from the left and right arguments of binary connectives, a locus of theoretical disagreement. We aimed to adjudicate between two camps of theories. On *c*-theories, presuppositions triggered in the left argument project unconditionally, becoming presuppositions of the entire sentence, while presuppositions triggered in the right argument project only conditionally. In contrast, *u*-theories posit that presuppositions triggered in either argument project unconditionally. A persistent difficulty in arbitrating between these approaches is that extralinguistic factors easily affect what these sentences are felt to presuppose, which can obscure introspective judgments about presuppositions and undermine their reliability. To address this, we leveraged cancellation costs, measured through response time increases, as an indirect method to assess the strength of presuppositions projected by binary connectives.

In our critical trials, participants evaluated whether an outcome complied with a statement (an instruction or a prediction) when the outcome failed to support, unconditionally, a presupposition triggered in the statement. Crucially, we manipulated whether the presupposition trigger originated in the left or right argument of the binary connective. Our primary question was whether an unmet presupposition’s location would lead to a measurable asymmetry in processing costs.

With both types of connectives, we found asymmetries in either judgments or in response times. In Experiments 1 and 3, participants were slower to accept an outcome as compliant when the unmet presupposition was triggered in the left argument. In Experiment 2, participants were less likely to judge outcomes as compliant with the presuppositional statement if the unmet presupposition was triggered in the left argument. These results are summarized in Table 19.

Table 19: Summary of findings for critical trials across experiments

Experiment	Connective	Yes-rates	Response Times
Exp 1	Conditional	left = right	left > right (*)
Exp 2	Conditional	left < right (*)	left > right (n.s)
Exp 3	Disjunction	left = right	left > right (*)

These asymmetries can be seen as providing support to *c*-theories, which predict differences in the strength of presuppositions projected by the two arguments: presuppositions projected from the left argument are stronger than those from the right. When the trigger is in the right argument, the presupposition is weaker and conditional, making it compatible with our critical contexts. When the trigger is in the left argument, on the other hand, contexts incompatible with the stronger presupposition could lead to more “No” responses. This also explains the response-time increase: when participants responded “Yes,” they would have needed to cancel the stronger presupposition only when it originated in the left argument, leading to a selective processing cost.

4.2 Alternative explanations in line with *u*-theories

The experimental findings are less straightforward to explain on *u*-theories, which which predict unconditional presuppositions for both trigger positions by default. By the logic of our experiments, this should lead to equally high rates of “No” responses or force cancellation at equal rates to make the outcomes in the critical cases compatible with the presuppositional sentences. Crucially, in the latter case, the processing costs should be comparable.

At the same time, it is not impossible for the *u*-theorists to find alternative explanations to our findings. One way to do so is to challenge our linking assumptions about the response time asymmetries. One of our critical assumptions is that the processing cost of canceling a presupposition is independent of trigger location. Future research could reveal independent reasons to doubt this assumption. If it would turn out that canceling presuppositions from left compared to right arguments incurs higher processing costs, that would provide a suitable alternative explanation. In this case asymmetry in *cost of cancellation* rather than asymmetry in *presence of cancellation* would be driving the reaction times differences.

Another option is that the asymmetry is caused by task-specific processing demands. In the experiments, participants first read the presuppositional sentences and then saw the outcome. If a presupposed item did not appear in the outcome, participants could have double-checked the sentence, causing the slowdown. It is not inconceivable that there was a recency advantage: the earlier the presupposition is encountered in the sentence, the greater the slowdown. Again, in this scenario, we would observe the same response time differences even if the strength of the presuppositions projected from the two arguments is the same. Both of these alternative explanations are speculative and require additional, yet to be attested assumptions about cancellation and processing respectively. Nevertheless both are empirically testable in future work and could provide support for *u*-theories.

4.3 Comparison to previous work

We are not the first to experimentally investigate presupposition projection from binary connectives. Here, we review earlier findings to situate our work and clarify our contributions.

Only two experimental studies have focused specifically on the proviso problem, with inconclusive results on whether symmetric or asymmetric theories are on the right track. Romoli et al. (2011) used a covered box task to probe interpretations of conditionals with presuppositions triggered in the consequent. Participants often selected images that verified the conditional presupposition, showing tentative support for *c*-theories theories, though selection rates were 10% lower when the antecedent and presupposition were unrelated (see e.g. van Rooij 2007, Lassiter 2012).

Domaneschi et al. (2016) used an interference task to probe whether conditional or unconditional presuppositions are more ‘basic’ for sentences of the form “If A, then B_p”. They found that the rates at which participants associated such sentences with a conditional presupposition decreased with cognitive load. The authors interpret this result as more in line with the *u*-theories, with cognitive load interfering with participants’ ability to weaken an inherently strong presupposition. Our results can be seen as coming down on the same side as Romoli et al. Using a different task and different behavioral measure, we find evidence that is best explained on *c*-theories.

Other studies experimentally investigating projection have focused on a different debate. A

subset of *c*-theories predict conditionalized presuppositions also from the left argument of a binary connective (Schlenker 2008, 2009; George 2008a, 2008b; Fox 2008; Rothschild 2008; Chemla 2009). A number of studies have tested this prediction experimentally⁹, and the results have been mixed. Some studies show that conditional presuppositions can obtain even when the presupposition is triggered in the first argument of a connective (e.g., Chemla & Schlenker 2012, Schwarz 2016a for conditionals; Kalomoiros & Schwarz 2021a, 2021b for disjunctions), while others (e.g., Mandelkern et al. 2020 for conjunctions) do not.

Our findings could be seen as providing indirect corroboration for the left argument having a stronger presupposition, if the observed asymmetries reflect the cost of asymmetric application of cancellation. But with a different linking hypothesis, our data could be made compatible with theories that predict weaker presuppositions from the left argument. Proponents of these theories have argued that the perceived left-right asymmetry is due to the fact that presuppositions are checked incrementally as the sentence unfolds in real-time, i.e., as soon as a trigger is encountered and against a context containing only information that preceded it. Though this constraint is violable, violating it comes at a cost. If this is correct, the asymmetric costs associated with the left argument need not be the cost of cancellation, but the cost of violating this constraint (see discussion in Kalomoiros 2023; see also Schlenker 2008, 2009; Chemla and Schlenker 2012.). Finally, we should be cautious when comparing this body of work to the present study. The core question addressed by these studies (the nature of projection from the *left* argument) is fundamentally different from ours, and they all rely on different experimental logic and measure from ours (typically acceptability judgments or rating robustness of inferences).

5 Conclusion

In a set of processing experiments with adults, we showed that participants respond differently to presuppositions triggered in the left and right arguments of binary connectives. We took these results to lend support for a family of theories on which only presuppositions triggered in the left argument become presuppositions of the sentence on the whole. Our study presents a further argument for using processing measures to inform how speakers interpret presuppositional sentences. The recruitment of presupposition-cancellation mechanisms is costly, and this cost is detectable in participants' response times even when intuitions are subtle or unstable.

Data availability

All data and analysis code of experiments reported in this paper are made available via the Open Science Framework (OSF) and can be accessed [here](#). Pre-registration for **Exp 1** and **Exp 2** are also hosted on OSF. Exp 3 was not pre-registered but we followed the same procedure and analysis plan as Exp 2, but in both Exp 2 and Exp 3, we deviated from the pre-registered analysis in two ways: (1) in analyzing Yes responses, we opted for mixed-effects logistic regression modeling, so as to account for between-participant variance, instead of planned chi-squared tests; (2) in analyzing

⁹Chemla & Schlenker (2012), Schwarz (2016a), Schwarz & Tiemann (2017), Hirsch & Hackl (2014), Hirsch, Zehr & Schwarz (2018), Mandelkern et al. (2020), Kalomoiros & Schwarz (2021a, 2021b), Kalomoiros 2023; for an overview, see Schwarz (2016b)

RTs, we fit a single model on all 3 trial types (p_true, p_critical, and control), whereas the planned analysis involved fitting separate models for p_true vs. p_critical and controls.

Ethics and Consent

All studies reported here received prior approval from the Institutional Review Board (IRB) under Massachusetts Institute of Technology Protocol 1303005593. Written consent was obtained from all participants, in accordance with ethical guidelines.

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A Appendix: Details on filler items

Experiments 1 and 2 Experiments 1 and 2 included two types of filler trials with “No” as the expected response. These items were also used as attention-checks and figured into identifying and excluding inattentive participants. A further role of the non-presuppositional false fillers was to mitigate against participants adopting image-based strategies when answering p_critical and control conditions. Like those conditions, the stick figure here has only one clothing item, but the

expected response is the opposite.

See below and Table 20.

- **p_false** (×12): the item described by the presupposition trigger is present in the dress-up; all requirements from the antecedent clause of the conditional are satisfied, rendering the antecedent true, but the requirements from the consequent are not satisfied.
- **non-presuppositional false** (×24): the sentence contains no presuppositions; items mentioned in antecedent is part of the outcome, but those mentioned in the consequent are not. this rendering the entire conditional false.

Table 20: False filler trial types in Experiments 1/2

Trial Type		Outcome Image	Statements
p_false	Antecedent		You have to give him an umbrella. You don't have to give him a pipe. But if the briefcase you give is black, you do have to give him a pipe.
	Consequent		You have to give him an umbrella. You don't have to give him a pipe. But if you do give him a pipe, the briefcase you give him has to be black.
Non-presuppositional false			You don't have to give him an umbrella. You don't have to give him a pipe. But if you do give him a pipe, you have to give him an umbrella.

Experiment 3 We included 4 types of fillers in Experiment 3. To balance “Yes” responses across the experiment, the majority of these fillers had a “No” expected response. These were used for participant exclusion.

- **p_false** (×8): item described by the embedded presupposition trigger appears in the outcome, but in the wrong color, making that disjunct false; the item mentioned in the non-presuppositional disjunct does not appear, making that disjunct false as well. Participants are expected to respond “No”.
- **NeitherNor** (×8): sentences of the form “He has neither *Object*₁ nor *Object*₂”, where a clothing item described in one or other disjunct appears in the outcome image. The true disjunct falsifies the sentence as a whole, so participants are expected to respond “No”.
- **Non-presuppositional disjunction, NeitherDisjunct** (×12): non-presuppositional disjunctive sentence; none of the mentioned clothing items appear in the dress-up, making the disjunction false. Participants are expected to respond “No”.
- **Non-presuppositional disjunction, BothDisjuncts** (×4): a non-presuppositional disjunctive sentence, where clothing items mentioned in both of the disjuncts appear in the dress-up. The disjunction is therefore true on an “inclusive” reading of *or*. Participants were trained to acceptable such sentences and thus are expected to respond “Yes”.